

Putting Machine Learning in Context

To understand the role of machine learning, we need to give you some context. AI, machine learning, and deep learning are all terms that are frequently mentioned when discussing big data, analytics, and advanced technology. AI can be understood as the

broadest way of describing systems that can “think.” For example, thermostats that learn your preference or applications that can identify people and what they are doing in photographs can be thought of as AI systems.

As illustrated in Figure 1-1, there are four main subsets of AI. In this book, we focus on machine learning. However, in order to understand machine learning, it is important to put it in perspective.

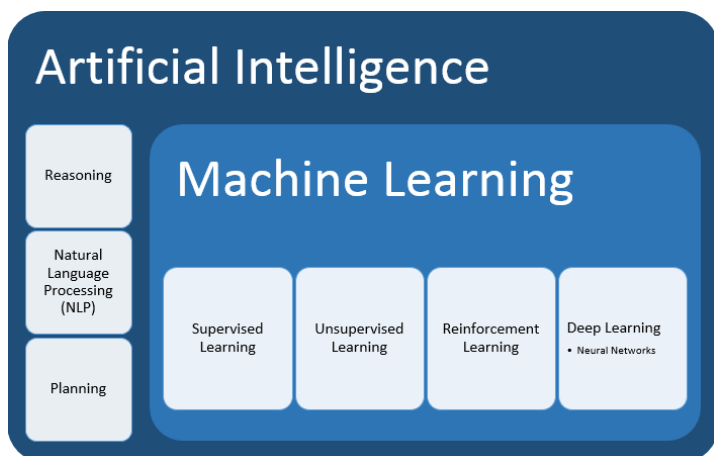


FIGURE 1-1: AI is the overall category that includes machine learning and natural language processing.



REMEMBER

When we explore machine learning, we focus on the ability to learn and adapt a model based on the data rather than explicit programming. In Chapter 6, we focus on applying machine learning to solving business problems.

Before we delve into the types of machine learning, it is important to understand the other subsets of AI:

» **Reasoning:** Machine reasoning allows a system to make inferences based on data. In essence, reasoning helps fill in the blanks when there is incomplete data. Machine reasoning helps make sense of connected data. For example, if a system has enough data and is asked “What is a safe internal temperature for eating a drumstick?” the system would be capable of telling you that the answer is 165 degrees. The

logic chain would be as follows: A drumstick that is eaten (as opposed to a part of a musical instrument) refers to a chicken leg, a chicken leg contains dark chicken meat, dark chicken meat needs to be cooked at 165 degrees, therefore the answer is 165 degrees. **Note:** In this example, the system was never explicitly trained on the safe internal temperature of chicken drumsticks. Instead the system used the knowledge it had to fill in the data gaps.

- » **Natural Language Processing (NLP):** NLP is the ability to train computers to understand both written text and human speech. NLP techniques are needed to capture the meaning of unstructured text from documents or communication from the user. Therefore, NLP is the primary way that systems can interpret text and spoken language. NLP is also one of the fundamental technologies that allows non-technical people to interact with advanced technologies. For example, rather than needing to code, NLP can help users ask a system questions about complex data sets. Unlike structured database information that relies on schemas to add context and meaning to the data, unstructured information must be parsed and tagged to find the meaning of the text. Tools required for NLP include categorization, ontologies, tapping, catalogs, dictionaries, and language models.
- » **Planning:** Automated planning is the ability for an intelligent system to act autonomously and flexibly to construct a sequence of actions to reach a final goal. Rather than a pre-programmed decision-making process that goes from A to B to C to reach a final output, automated planning is complex and requires a system to adapt based on the context surrounding the given challenge.

Approaches to Machine Learning

Machine learning techniques are required to improve the accuracy of predictive models. Depending on the nature of the business problem being addressed, there are different approaches based on the type and volume of the data. In this section, we discuss the categories of machine learning.

Supervised learning

Supervised learning typically begins with an established set of data and a certain understanding of how that data is classified. Supervised learning is intended to find patterns in data that can be applied to an analytics process. This data has labeled features that define the meaning of data. For example, there could be millions of images of animals and include an explanation of what each animal is and then you can create a machine learning application that distinguishes one animal from another. By labeling this data about types of animals, you may have hundreds of categories of different species. Because the attributes and the meaning of the data have been identified, it is well understood by the users that are training the modeled data so that it fits the details of the labels. When the label is continuous, it is a regression; when the data comes from a finite set of values, it is known as *classification*. In essence, regression used for supervised learning helps you understand the correlation between variables. An example of supervised learning is weather forecasting. By using regression analysis, weather forecasting takes into account known historical weather patterns and the current conditions to provide a prediction on the weather.



TIP

The algorithms are trained using preprocessed examples, and at this point, the performance of the algorithms is evaluated with test data. Occasionally, patterns that are identified in a subset of the data can't be detected in the larger population of data. If the model is fit to only represent the patterns that exist in the training subset, you create a problem called *overfitting*. Overfitting means that your model is precisely tuned for your training data but may not be applicable for large sets of unknown data. To protect against overfitting, testing needs to be done against unforeseen or unknown labeled data. Using unforeseen data for the test set can help you evaluate the accuracy of the model in predicting outcomes and results. Supervised training models have broad applicability to a variety of business problems, including fraud detection, recommendation solutions, speech recognition, or risk analysis.

Unsupervised learning

Unsupervised learning is best suited when the problem requires a massive amount of data that is unlabeled. For example, social media applications, such as Twitter, Instagram, Snapchat, and

so on all have large amounts of unlabeled data. Understanding the meaning behind this data requires algorithms that can begin to understand the meaning based on being able to classify the data based on the patterns or clusters it finds. Therefore, the supervised learning conducts an iterative process of analyzing data without human intervention. Unsupervised learning is used with email spam-detecting technology. There are far too many variables in legitimate and spam emails for an analyst to flag unsolicited bulk email. Instead, machine learning classifiers based on clustering and association are applied in order to identify unwanted email.



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Unsupervised learning algorithms segment data into groups of examples (clusters) or groups of features. The unlabeled data creates the parameter values and classification of the data. In essence, this process adds labels to the data so that it becomes supervised. Unsupervised learning can determine the outcome when there is a massive amount of data. In this case, the developer doesn't know the context of the data being analyzed, so labeling isn't possible at this stage. Therefore, unsupervised learning can be used as the first step before passing the data to a supervised learning process.



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Unsupervised learning algorithms can help businesses understand large volumes of new, unlabeled data. Similarly to supervised learning (see the preceding section), these algorithms look for patterns in the data; however, the difference is that the data is not already understood. For example, in healthcare, collecting huge amounts of data about a specific disease can help practitioners gain insights into the patterns of symptoms and relate those to outcomes from patients. It would take too much time to label all the data sources associated with a disease such as diabetes. Therefore, an unsupervised learning approach can help determine outcomes more quickly than a supervised learning approach.

Reinforcement learning

Reinforcement learning is a behavioral learning model. The algorithm receives feedback from the analysis of the data so the user is guided to the best outcome. Reinforcement learning differs from other types of supervised learning because the system isn't trained with the sample data set. Rather, the system learns through trial and error. Therefore, a sequence of successful decisions will result in the process being "reinforced" because it best solves the problem at hand.



One of the most common applications of reinforcement learning is in robotics or game playing. Take the example of the need to train a robot to navigate a set of stairs. The robot changes its approach to navigating the terrain based on the outcome of its actions. When the robot falls, the data is recalibrated so the steps are navigated differently until the robot is trained by trial and error to understand how to climb stairs. In other words, the robot learns based on a successful sequence of actions. The learning algorithm has to be able to discover an association between the goal of climbing stairs successfully without falling and the sequence of events that lead to the outcome.

Reinforcement learning is also the algorithm that is being used for self-driving cars. In many ways, training a self-driving car is incredibly complex because there are so many potential obstacles. If all the cars on the road were autonomous, trial and error would be easier to overcome. However, in the real world, human drivers can often be unpredictable. Even with this complex scenario, the algorithm can be optimized over time to find ways to adapt to the state where actions are rewarded. One of the easiest ways to think about reinforcement learning is the way an animal is trained to take actions based on rewards. If the dog gets a treat every time he sits on command, he will take this action each time.

Neural networks and deep learning

Deep learning is a specific method of machine learning that incorporates neural networks in successive layers in order to learn from data in an iterative manner. Deep learning is especially useful when you're trying to learn patterns from unstructured data.

Deep learning — complex neural networks — are designed to emulate how the human brain works so computers can be trained to deal with abstractions and problems that are poorly defined. The average five-year-old child can easily recognize the difference between his teacher's face and the face of the crossing guard. In contrast, the computer has to do a lot of work to figure out who is who. Neural networks and deep learning are often used in image recognition, speech, and computer vision applications.

A neural network consists of three or more layers: an input layer, one or many hidden layers, and an output layer. Data is ingested through the input layer. Then the data is modified in the hidden layer and the output layers based on the weights applied to

these nodes. The typical neural network may consist of thousands or even millions of simple processing nodes that are densely interconnected. The term deep learning is used when there are multiple hidden layers within a neural network. Using an iterative approach, a neural network continuously adjusts and makes inferences until a specific stopping point is reached. Neural networks are often used for image recognition and computer vision applications.

Deep learning is a machine learning technique that uses hierarchical neural networks to learn from a combination of unsupervised and supervised algorithms. Deep learning is often called a sub-discipline of machine learning. Typically, deep learning learns from unlabeled and unstructured data. While deep learning is very similar to a traditional neural network, it will have many more hidden layers. The more complex the problem, the more hidden layers there will be in the model.



REMEMBER

There are many areas where deep learning will have an impact on businesses. For example, voice recognition will have applications in everything from automobiles to customer management. In the Internet of Things (IoT) manufacturing applications, deep learning can be used to predict when a machine will malfunction. Deep learning algorithms can help law enforcement personnel keep track of the movements of a known suspect.